Comprehensive Matrix and Linear Algebra Overview

Your Name

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Outline (Slide 1)

Outline (Slide 2)

5) Matrix Multiplication

- $n \times m$ to $m \times k$ multiplication
- Visual examples
- Matrix-vector multiplication ($n \times m$ on $m \times 1$)
- 6×4 on 4×1
- 3×3 example
- NumPy (dot, @)
- Identity multiplication

6) Transpose Matrix

- $n \times m$ matrices
- 4×3 example
- NumPy examples

7) Inverse Matrix

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- 2×2, 3×3, $n \times m$ examples
- Inverse matrix
- Visual examples
- NumPy examples

Outline (Slide 3)

Outline (Slide 4)

Matrix Definition

- A **matrix** is a rectangular array of numbers organized in rows and columns.
- Example of a 2×3 matrix:

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}.$$

Matrix Dimensions in NumPy

```
import numpy as np

A = np.array([[1, 2, 3],
              [4, 5, 6]])
print(A.shape)  # Output: (2, 3)
```

Examples of Various Dimensions

$$2 \times 2: \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

$$3 \times 3: \quad \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix},$$

$$2 \times 3: \quad \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix},$$

$$3 \times 2: \quad \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}.$$

Visual Matrix Examples (TikZ)

1	2	3
4	5	6

1	2
3	4
5	6

$n \times m$ Matrices

General form:

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix}.$$

Square Matrices

- A matrix is **square** if it has the same number of rows and columns ($n \times n$).
- Examples:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

NumPy: Various Matrix Shapes

```
import numpy as np

A_2x2 = np.array([[1,2],
                  [3,4]])
print(A_2x2.shape)  # Output: (2, 2)

A_3x3 = np.array([[1,2,3],
                  [4,5,6],
                  [7,8,9]])
print(A_3x3.shape)  # Output: (3, 3)

A_2x3 = np.array([[1,2,3],
                  [4,5,6]])
print(A_2x3.shape)  # Output: (2, 3)

A_3x2 = np.array([[1,2],
                  [3,4],
                  [5,6]])
```

Sum of Matrices (2×3 Example)

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} + \begin{pmatrix} 7 & 8 & 9 \\ 0 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 8 & 10 & 12 \\ 4 & 6 & 8 \end{pmatrix}.$$

Subtraction of Matrices (3×2 Example)

$$\begin{pmatrix} 4 & 5 \\ 2 & 3 \\ 1 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 1 \\ 2 & 2 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} 3 & 4 \\ 0 & 1 \\ 1 & -3 \end{pmatrix}.$$

Scalar Multiplication and Distributive Law (4×3 Example)

- Multiply each entry by a scalar α.
- **Distributive Law**: $\alpha(A+B) = \alpha A + \alpha B$.

$$2 \cdot \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \\ j & k & l \end{pmatrix} = \begin{pmatrix} 2a & 2b & 2c \\ 2d & 2e & 2f \\ 2g & 2h & 2i \\ 2j & 2k & 2l \end{pmatrix}.$$

Diagonal Matrix

- A **diagonal matrix** is a square matrix in which the entries outside the main diagonal are all zero.
- Formally, a matrix A is diagonal if:

$$A = \begin{pmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{pmatrix}.$$

- **Example of a 3×3 Diagonal Matrix**:

$$D = \begin{pmatrix} 5 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

- **Properties**:
  - **Multiplicative**: $D_1 \times D_2$ is also a diagonal matrix.
  - **Inverse**: If all diagonal entries are non-zero, D is invertible, and D^{-1} is obtained by taking reciprocals of the diagonal entries.
  - **Transpose**: $D^T = D$ (since it's symmetric).

Visual Example of a Diagonal Matrix

5	0	0
0	-3	0
0	0	2

Identity Matrix

- The identity matrix I_n is an $n \times n$ matrix with 1's on the main diagonal and 0's elsewhere.
- Example of a 4×4 identity matrix:

$$I_4 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

- For any $n \times n$ matrix A, $I_n A = A I_n = A$.

$(n \times m)$ Times $(m \times k)$

$$(A_{n\times m}) \times (B_{m\times k}) = C_{n\times k}.$$

- The inner dimensions (m) must match.
- The result has outer dimensions $n \times k$.

Visual Example of Multiplication

$A_{2\times 3}$

a	b	c
d	e	f

$\times$

$B_{3\times 2}$

p	q
r	s
t	u

$=$

$C_{2\times 2}$

A_{11}	A_{12}
A_{21}	A_{22}

Matrix-Vector Multiplication ($n \times m$ on $m \times 1$)

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \cdot x_1 + 2 \cdot x_2 + 3 \cdot x_3 \\ 4 \cdot x_1 + 5 \cdot x_2 + 6 \cdot x_3 \end{pmatrix}.$$

6×4 on 4×1 Vector Example

$$A_{6\times 4} \cdot \vec{x}_{4\times 1} = \vec{y}_{6\times 1}.$$

- Example matrix and vector:

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \\ 17 & 18 & 19 & 20 \\ 21 & 22 & 23 & 24 \end{pmatrix}, \quad \vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}.$$

- The result $\vec{y}$ is a 6×1 vector.

3×3 Multiplication Example

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \times \begin{pmatrix} p & q & r \\ s & t & u \\ v & w & x \end{pmatrix} =$$

$$\begin{pmatrix} ap + bs + cv & aq + bt + cw & ar + bu + cx \\ dp + es + fv & dq + et + fw & dr + eu + fx \\ gp + hs + iv & gq + ht + iw & gr + hu + ix \end{pmatrix}$$

NumPy Matrix Multiplication

```
import numpy as np

A = np.array([[1,2,3],
              [4,5,6]])
B = np.array([[7,8],
              [9,10],
              [11,12]])

C1 = A.dot(B)
C2 = A @ B  # Python 3.5+ operator
print("C1:\n", C1)
print("C2:\n", C2)
```

Multiplication with Identity

$$I_n A = A I_n = A \quad (\text{if } A \text{ is } n \times n).$$

- Multiplying any square matrix by the identity matrix leaves it unchanged.

Transpose

$$(A^T)_{ij} = A_{ji}.$$

- Transposes an $n \times m$ matrix into an $m \times n$ matrix.

4×3 Transpose Example

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 0 & 1 & 2 \end{pmatrix} \quad \Rightarrow \quad A^T = \begin{pmatrix} 1 & 4 & 7 & 0 \\ 2 & 5 & 8 & 1 \\ 3 & 6 & 9 & 2 \end{pmatrix}.$$

Transpose in NumPy

```
import numpy as np

A = np.array([[1,2,3],
              [4,5,6],
              [7,8,9],
              [0,1,2]])
A_T = A.T
print("A Transposed:\n", A_T)
```

Matrix Determinant

- For a square matrix, the determinant $\det(A)$ determines if A is invertible.
- A is invertible if $\det(A) \neq 0$.

2×2 Determinant and Inverse

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

$$\text{If } ad - bc \neq 0, \text{ then } A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

3×3 Determinant

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg).$$

$n \times m$ Matrices

- Non-square matrices ($n \neq m$) do not have a standard inverse.
- Only square matrices ($n = m$) can potentially be invertible.

Inverse Matrix

$$A^{-1}A = AA^{-1} = I_n \quad \text{(if } A \text{ is } n \times n \text{ and non-singular).}$$

- Example for 2×2 matrix:

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Inverse Matrix Visual (2×2)

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix}$$

Inverse in NumPy

```
import numpy as np

A = np.array([[1.,2.],
              [3.,4.]])
A_inv = np.linalg.inv(A)
print("A Inverse:\n", A_inv)
```

Matrix Algebra in NumPy

```
import numpy as np

# Define matrices
A = np.random.randn(3,3)
B = np.random.randn(3,3)

# Operations
C = A + B          # Sum
D = A - B          # Subtraction
E = A.dot(B)       # Multiplication
F = np.linalg.inv(A)  # Inverse (if A is invertible)

print("Sum:\n", C)
print("Subtraction:\n", D)
print("Multiplication:\n", E)
print("Inverse of A:\n", F)
```

2D and 3D Matrices in NumPy

```
import numpy as np

# 2D array
A_2d = np.array([[1,2,3],
                 [4,5,6]])
print("2D shape:", A_2d.shape)  # Output: (2, 3)

# 3D array (e.g., RGB image)
A_3d = np.random.randn(2,3,4)  # Shape: (2,3,4)
print("3D shape:", A_3d.shape)
```

Tensors in NumPy

```
import numpy as np

# 3D tensor
T = np.random.randn(3,3,3)
print("Tensor shape:", T.shape)  # Output: (3,3,3)

# Higher-dimensional tensor
T4 = np.random.randn(2,3,4,5)
print("4D Tensor shape:", T4.shape)
```

Image as Matrix

- **Grayscale Image**: 2D matrix of pixel intensities.
- **Color Image (RGB)**: 3D tensor (height $\times$ width $\times$ channels).

Text as Matrix

- **Embedding**: Each word is represented as a vector, forming a matrix (sequence length $\times$ embedding dimension).
- Used in Natural Language Processing (NLP) for various applications.

Linear Independence and Dimension

- Vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ are **linearly independent** if:

$$\alpha_1 \mathbf{v}_1 + \cdots + \alpha_k \mathbf{v}_k = \mathbf{0} \quad \Longrightarrow \quad \alpha_1 = \cdots = \alpha_k = 0.$$

- A basis for an n-dimensional space has exactly n linearly independent vectors.

Basis Vectors - Definition

Definition: Basis vectors are a set of vectors in a vector space that are:

- **Linearly Independent.**
- **Span the Vector Space.**

Standard Basis Vectors in $\mathbb{R}^n$:

- In $\mathbb{R}^2$: $e_1 = [1, 0]^T$, $e_2 = [0, 1]^T$.
- In $\mathbb{R}^3$: $e_1 = [1, 0, 0]^T$, $e_2 = [0, 1, 0]^T$, $e_3 = [0, 0, 1]^T$.

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- In $\mathbb{R}^3$: $e_1 = [1, 0, 0]^T$, $e_2 = [0, 1, 0]^T$, $e_3 = [0, 0, 1]^T$.

Properties:

- Any vector can be written as a linear combination of basis vectors.
- Basis vectors define coordinate axes.

Linear Combinations

Definition: A vector $\mathbf{v}$ can be written as:

$$\mathbf{v} = c_1\mathbf{e}_1 + c_2\mathbf{e}_2 + \cdots + c_n\mathbf{e}_n$$

Examples:

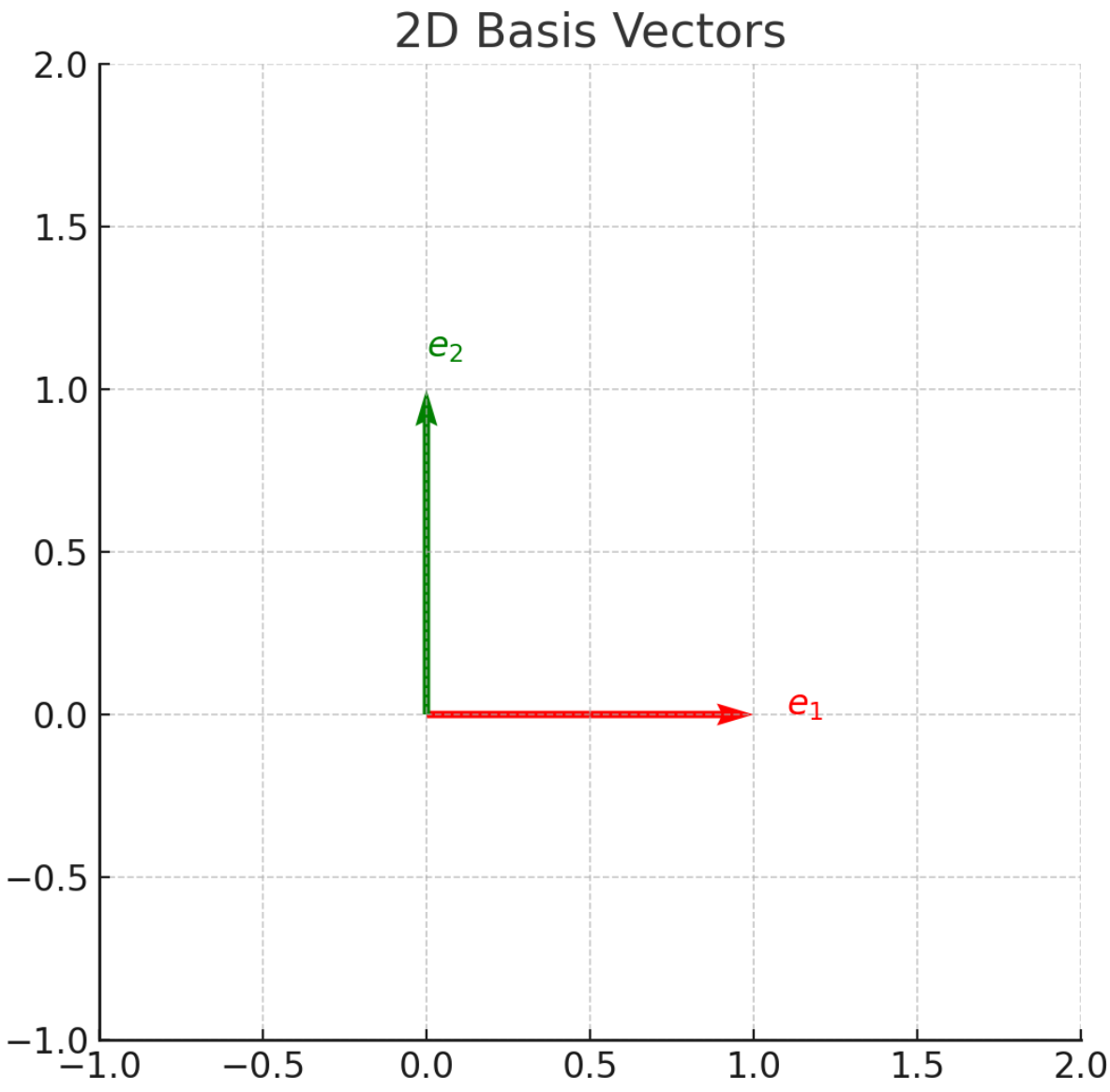
- In $\mathbb{R}^2$:

$$\mathbf{u} = [3, 4] = 3[1, 0] + 4[0, 1]$$

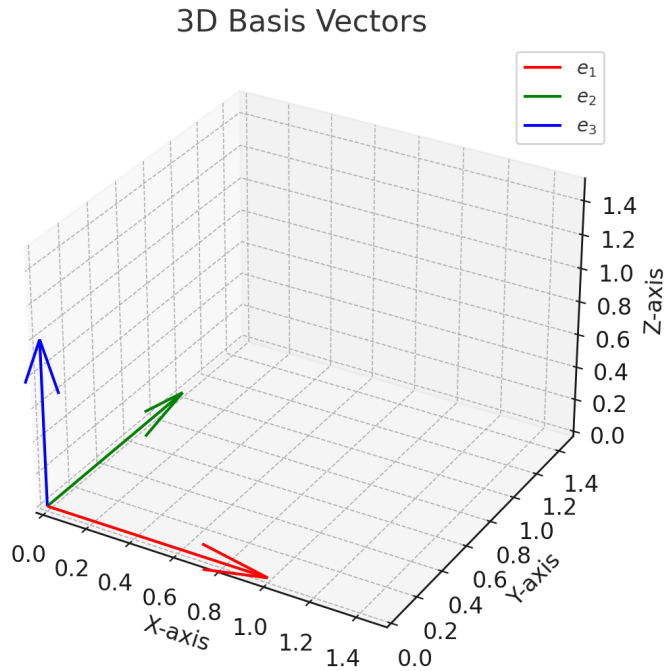
- In $\mathbb{R}^3$:

$$\mathbf{v} = [1, 2, 3] = 1[1, 0, 0] + 2[0, 1, 0] + 3[0, 0, 1]$$

Visual Example: Basis Vectors in 2D



Visual Example: Basis Vectors in 3D



Example: Decomposing vector $\mathbf{v} = [1, 2, 3]$ into basis vectors in 3D.

Exercises

Task:

- Express the following vectors as linear combinations of basis vectors:
  1. $\mathbf{w} = [2, -1]$ in $\mathbb{R}^2$.
  2. $\mathbf{x} = [1, 0, -2]$ in $\mathbb{R}^3$.

Exercises

Task:

- Express the following vectors as linear combinations of basis vectors:
  1. $\mathbf{w} = [2, -1]$ in $\mathbb{R}^2$.
  2. $\mathbf{x} = [1, 0, -2]$ in $\mathbb{R}^3$.
- Verify your answers by reconstructing the vectors.

Key Takeaways

Key Concepts:

- Basis vectors define the axes in vector spaces.
- Any vector can be represented as a linear combination of basis vectors.
- Visualization helps in understanding the decomposition into components.

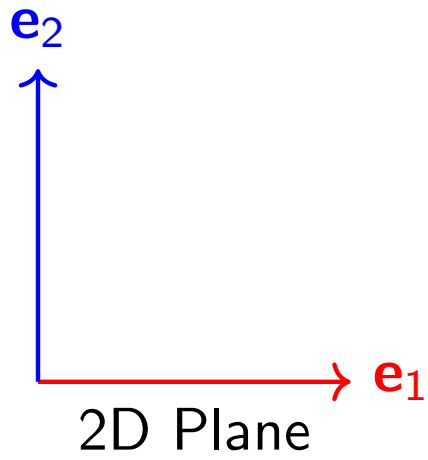
Next Steps: Apply these concepts to higher-dimensional spaces and affine transformations.

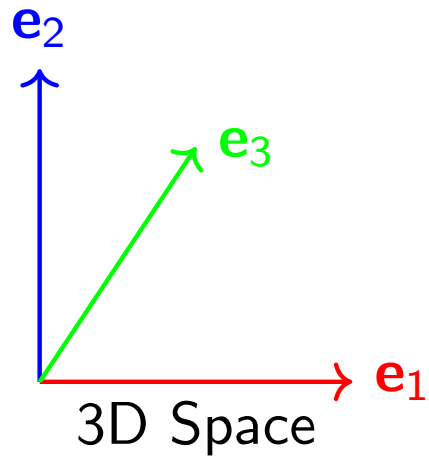
2D and 3D Space Examples

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \ \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (2D)$$

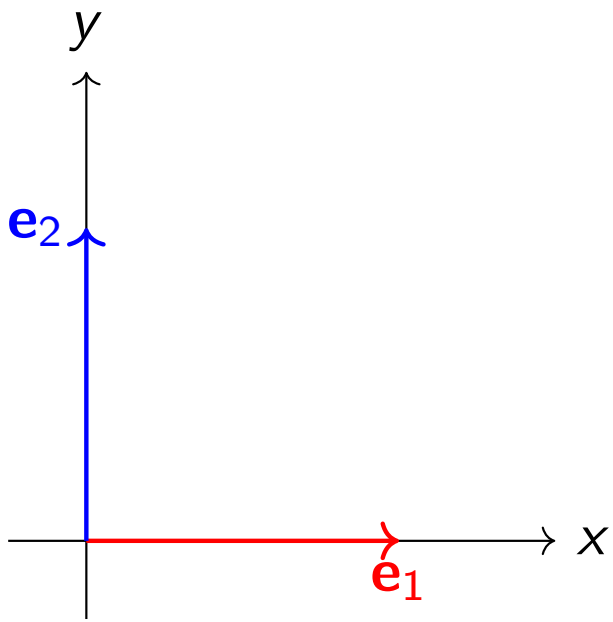
$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \ \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \ \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad (3D)$$

Visual Example of Basis Vectors





Basis in 2D Space



Checking Linear Independence in NumPy

```
import numpy as np

v1 = np.array([1,2,3])
v2 = np.array([2,4,6])
# v2 is multiple of v1 => dependent

M = np.column_stack((v1, v2))
rank = np.linalg.matrix_rank(M)
print("Rank =", rank)  # Output: Rank = 1
```

Matrix from Basis Vectors

- If $T(\mathbf{e}_i) =$ the i-th column of A, then for any $\mathbf{x}$,

$$T(\mathbf{x}) = A\mathbf{x}.$$

Proof Sketch

- A linear map $T : \mathbb{R}^n \to \mathbb{R}^m$ is fully determined by $T(\mathbf{e}_1), \ldots, T(\mathbf{e}_n)$.
- Each $T(\mathbf{e}_i)$ becomes the i-th column of matrix A representing T.

Linear Transformation in NumPy

```
import numpy as np

A = np.array([[2,1],
              [0,3]])
x = np.array([1,4])

Tx = A @ x
print("T(x) =", Tx)  # Output: T(x) = [6 12]
```

Composition of Linear Transformations

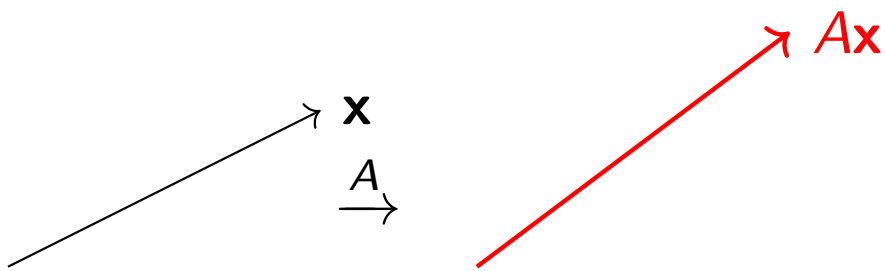
$$T_2 \circ T_1(\mathbf{x}) = A_2(A_1\mathbf{x}) = (A_2A_1)\mathbf{x}.$$

- Example:

$$A_1 \text{ is } 2 \times 3, \quad A_2 \text{ is } 3 \times 4 \quad \Rightarrow \quad A_2A_1 \text{ is } 2 \times 4.$$

- Ensures dimensional compatibility for matrix multiplication.

Visual Example of $T(\mathbf{x})$



Introduction to Linear Transformations

- A **linear transformation** is a mapping $T : \mathbb{R}^n \to \mathbb{R}^m$ that satisfies:
  1. **Additivity**: $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$
  2. **Homogeneity**: $T(c\mathbf{u}) = cT(\mathbf{u})$
- Every linear transformation can be represented by a matrix.
- Understanding how to generate these matrices is crucial for applications in computer graphics, data science, and more.

Generating Transformation Matrices

- To generate a linear transformation matrix A, determine how the transformation T maps each of the standard basis vectors $\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n$.
- The images $T(\mathbf{e}_1), T(\mathbf{e}_2), \ldots, T(\mathbf{e}_n)$ become the **columns** of matrix A.
- Mathematically:

$$A = \begin{pmatrix} | & | & & | \\ T(\mathbf{e}_1) & T(\mathbf{e}_2) & \ldots & T(\mathbf{e}_n) \\ | & | & & | \end{pmatrix} = [T(\mathbf{e}_1)\ T(\mathbf{e}_2)\ \ldots\ T(\mathbf{e}_n)]$$

- This ensures that for any vector $\mathbf{x} \in \mathbb{R}^n$,

$$T(\mathbf{x}) = A\mathbf{x}.$$

Scaling Transformation

- **Definition**: Scaling stretches or shrinks vectors by a constant factor along each axis.
- **Scaling Factors**: s_x along the x-axis and s_y along the y-axis.

$$A = \begin{pmatrix} s_x & 0 \\ 0 & s_y \end{pmatrix}$$

- **Example**: Scale by 2 along x-axis and 3 along y-axis.

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

Rotation Transformation

- **Definition**: Rotates vectors by an angle θ counterclockwise around the origin.
- **Rotation Matrix**:

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

- **Example**: Rotate by 45 degrees ($\theta = \frac{\pi}{4}$).

$$A = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

Shearing Transformation

- **Definition**: Shearing shifts each point in a fixed direction, proportional to its distance from a line that is parallel to that direction.
- **Types of Shear**:
  1. **Horizontal Shear**: Shifts x-coordinate based on y.
  2. **Vertical Shear**: Shifts y-coordinate based on x.
- **Shear Matrices**:

$$\text{Horizontal Shear: } A = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}, \quad \text{Vertical Shear: } A = \begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$$

Example: Horizontal shear with $k = 1.5$

$$A = \begin{pmatrix} 1 & 1.5 \\ 0 & 1 \end{pmatrix}$$

Reflection Transformation

- **Definition**: Reflects vectors across a specified axis or line.
- **Reflection Across the x-axis**:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- **Reflection Across the y-axis**:

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

- **Reflection Across the Line $y = x$**:

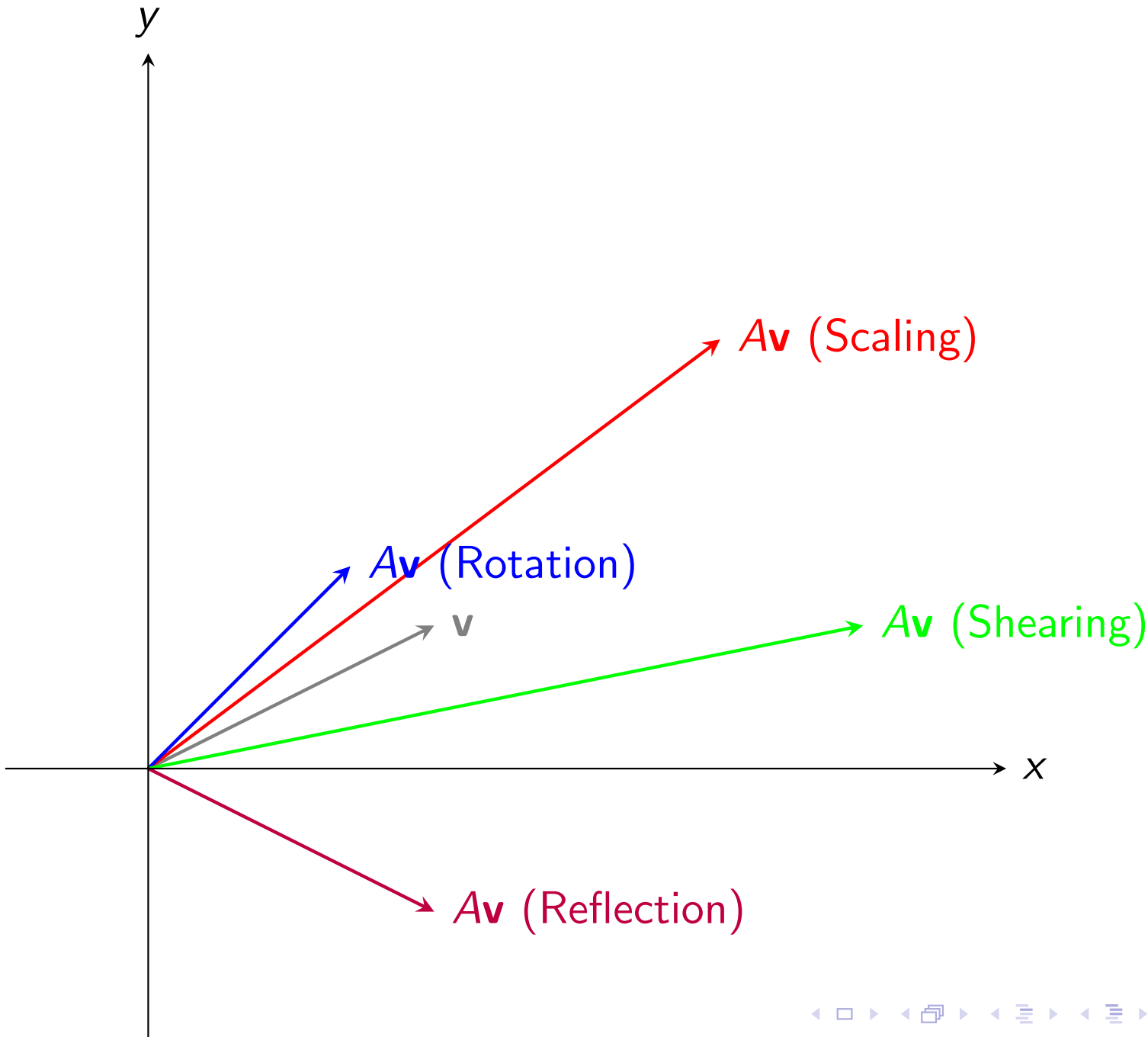
$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Generating Transformation Matrices in NumPy

- Use NumPy to create transformation matrices programmatically.
- Examples include scaling, rotation, shearing, and reflection.

```
import numpy as np

# Scaling Transformation
def scaling_matrix(sx, sy):
    return np.array([[sx, 0],
                     [0, sy]])

# Rotation Transformation
def rotation_matrix(theta):
    return np.array([[np.cos(theta), -np.sin(theta)],
                     [np.sin(theta),  np.cos(theta)]])

# Shearing Transformation (Horizontal)
def shear_horizontal(k):
    return np.array([[1, k]
```

Visual Examples of Linear Transformations



Affine Transformation between $\mathbb{R}^m$ and $\mathbb{R}^n$

- An **affine transformation** is a function $T : \mathbb{R}^m \to \mathbb{R}^n$ that can be expressed as:

$$T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$$

  - A is an $n \times m$ **transformation matrix**.
  - $\mathbf{b}$ is a fixed vector in $\mathbb{R}^n$ known as the **translation vector**.
- Affine transformations combine **linear transformations** with **translations**.
- Unlike linear transformations, affine transformations do not necessarily preserve the origin unless $\mathbf{b} = \mathbf{0}$.

Properties of Affine Transformations

- **Combination of Operations**:
  - Affine transformations are combinations of linear transformations and translations.
  - Formally, $T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ where A is linear and $\mathbf{b}$ is a translation.
- **Closure**:
  - The set of affine transformations is closed under composition.
- **Preservation of Parallelism**:
  - Affine transformations preserve parallel lines.
- **Linear Transformation as a Special Case**:
  - When $\mathbf{b} = \mathbf{0}$, the affine transformation reduces to a linear transformation.

Affine Transformation

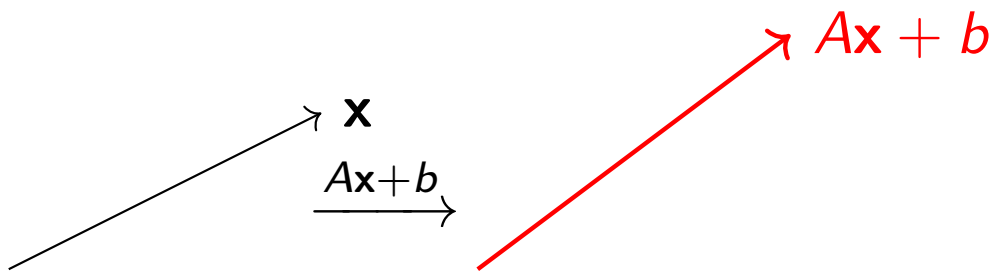
```
import numpy as np

A = np.array([[2,1],[0,3]])
b = np.array([1, -1])
x = np.array([3,2])

y = A @ x + b
print("Affine transform =", y)  # Output: Affine transform
```

Composition of Affine Transformations

- $f_1(\mathbf{x}) = A_1\mathbf{x} + b_1$
- $f_2(\mathbf{y}) = A_2\mathbf{y} + b_2$
- Composition:

$$f_2 \circ f_1(\mathbf{x}) = A_2(A_1\mathbf{x} + b_1) + b_2 = A_2A_1\mathbf{x} + (A_2b_1 + b_2).$$

Affine Transformation Visual



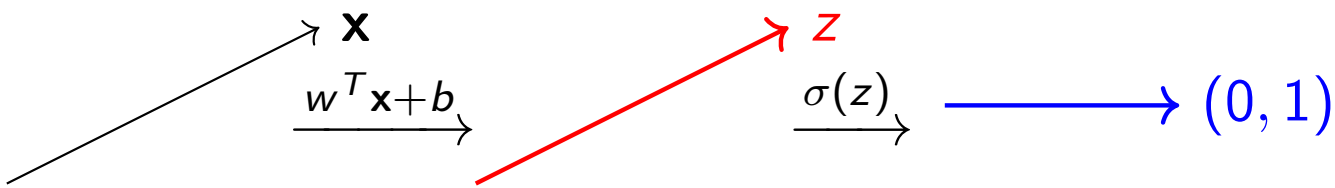
Transformation Formulas

$$f(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + b, \quad \mathbf{w} \in \mathbb{R}^n,\ b \in \mathbb{R}.$$

Affine Transformation with Sigmoid

```
import numpy as np

def sigmoid(z):
    return 1 / (1 + np.exp(-z))

w = np.array([1.5, -2.0])
b = 0.5
x = np.array([2.0, 1.0])

z = w @ x + b
y = sigmoid(z)
print("Affine + Sigmoid =", y)  # Output: Affine + Sigmoid
```

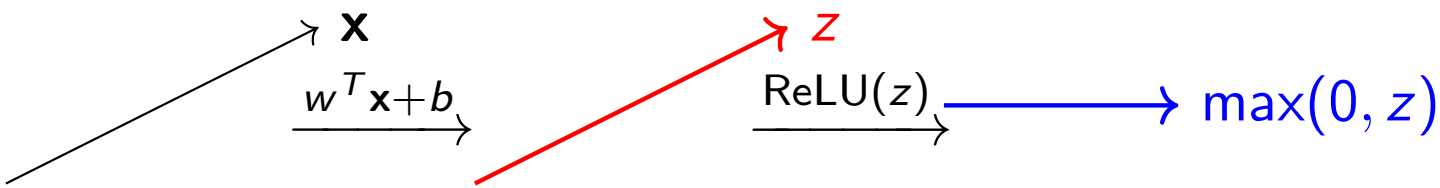
Visual: Affine $\rightarrow$ Sigmoid



NumPy Implementation

```
import numpy as np

def sigmoid(z):
    return 1 / (1 + np.exp(-z))

x = np.array([0.5, 2.0])
z = w @ x + b
output = sigmoid(z)
print("Output:", output)  # Output: Output: 0.880797077977
```

Affine + ReLU (Visual)



ReLU in NumPy

```
import numpy as np

def relu(z):
    return np.maximum(0, z)

w = np.array([1., -1.])
b = -0.5
x = np.array([2., 3.])

z = w @ x + b
y = relu(z)
print("Affine + ReLU =", y)  # Output: Affine + ReLU = 1.5
```